

What is your counterparty's promise to pay worth?

CVA and DVA with simulation and *machine learning*

Maurilio Patiño García

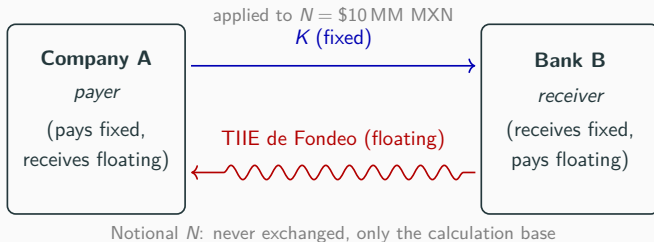
Universidad Anahuac
Master's in Risk

2008: CVA enters derivatives valuation

According to the Basel Committee, roughly one third of counterparty risk losses in the crisis came from defaults; the other two thirds came from the deterioration in the market value of the counterparties' promise to pay. (quoted in Gregory, 2012, sec. 17.4.4)

- A swap with a AAA bank and the same swap with a fragile counterparty **are not worth the same**. That discount has a name: **CVA**.
- It is now part of fair value measurement (IFRS 13; in Mexico, NIF B-17 and C-10), and Banco de Mexico requires the adjustments monthly in its **XVA report**.
- That is why many banks run **dedicated CVA desks**: teams that compute and hedge this number.

The instrument: an interest rate swap (IRS)



The *swap* in this class: fixed *payer*, 5 years, 10 MM MXN notional, TIIE de Fondeo, *at-the-money*. MexDer launched its TIIE de Fondeo swap on 26 February 2024; CME has cleared OIS on this rate since 5 February 2024.

The whole idea in one formula

$$\text{CVA} = \underbrace{(1 - R)}_{\text{severity}} \int_0^T \underbrace{\text{DF}(t) \text{EE}(t)}_{\text{discounted exposure}} \underbrace{d\text{PD}(t)}_{\text{probability}}$$

Severity

If it defaults, how much is lost?

$R = 0.25$ (EM CDS convention)

Exposure

How much will it owe me at each future date?

We must simulate

Probability

How likely is it to default?

Read off the CDS market

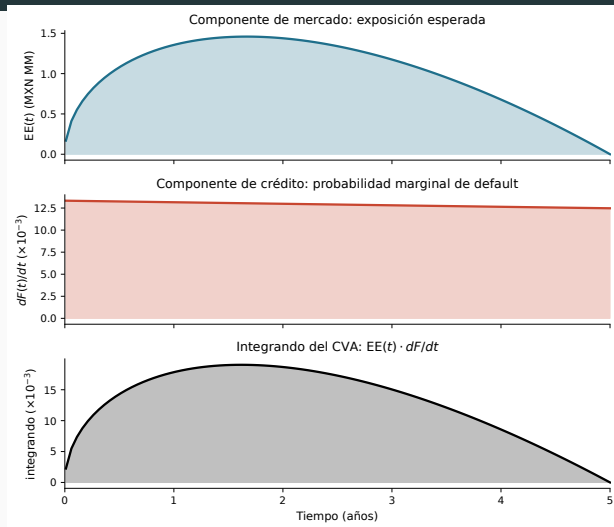
Why market probabilities and not historical ones? Because CVA is a **price** (hedged by buying CDS), not an actuarial forecast: risk-neutral valuation, under the measure $\mathbb{Q}$.

Assumptions: exposure independent of default. The *swap* is valued in each scenario with its rates; exposure is brought to today with the market curve (+0.25% in the thesis).

How the integrand is built (conceptual sketch)

- Top: expected exposure, $EE(t)$.
- Middle: marginal default probability.
- Bottom: their product. With discounting included, its area times $(1 - R)$ is the CVA.

Conceptual example from the thesis (ch. 2),
on a different scale from the reference *swap*.

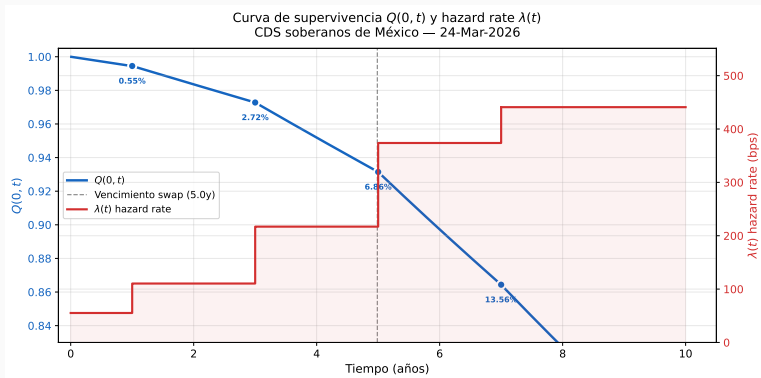


The mirror image: DVA

$$\text{DVA} = (1 - R_{\text{own}}) \int_0^T \text{DF}(t) \text{ENE}(t) d\text{PD}_{\text{own}}(t)$$

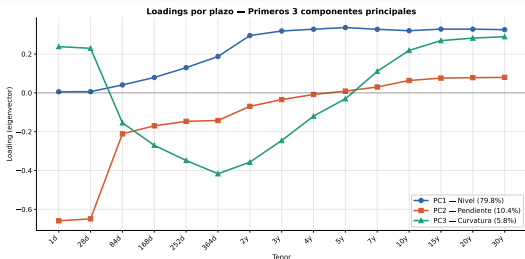
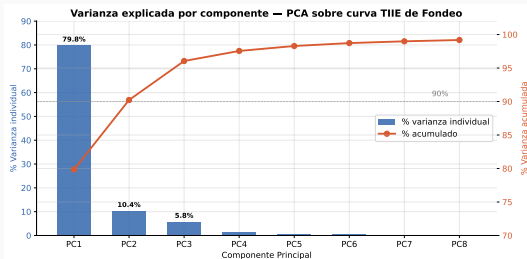
- The **same formula** seen from the other side of the table: the *negative* exposure with *our own* credit risk.
- With the same models and data, your DVA is the CVA your counterparty computes on you.
- Subtracting CVA from DVA gives a **simplified bilateral adjustment**: it does not model which party defaults first (Gregory, 2012).
- At the end of the class we compute it live, in three lines of code.

Piece 1: the probability is read off the market



- The CDS *spread* is the price of credit protection; a *bootstrap* turns it into a survival curve: $\lambda \approx s/(1 - R)$.

Piece 2: exposure requires simulating the future

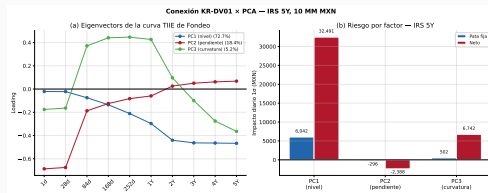


Level (79.85%) + slope (10.38%) = 90.23% of the variance. With one factor, every tenor depends on a single source of uncertainty; with two, level and slope can move separately.

The model: G2++, two correlated Gaussian factors

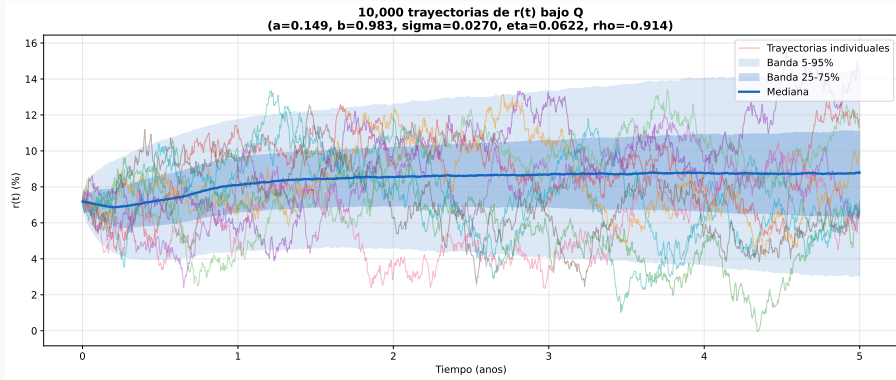
$$r(t) = x(t) + y(t) + \varphi(t),$$
$$\begin{aligned} dx(t) &= -a x(t) dt + \sigma dW_1(t) \\ dy(t) &= -b y(t) dt + \eta dW_2(t) \end{aligned}$$
$$dW_1 dW_2 = \rho dt$$

- **Slow** factor ($a = 0.149$): persists; associated with the **level**.
- **Fast** factor ($b = 0.983$): dies out; associated with the **slope**.
- $\rho = -0.914$: correlated factors; their link to PC1 and PC2 is approximate.
- ATM *swaptions* (Bloomberg, 24-Mar-2026); simulation exact per factor.

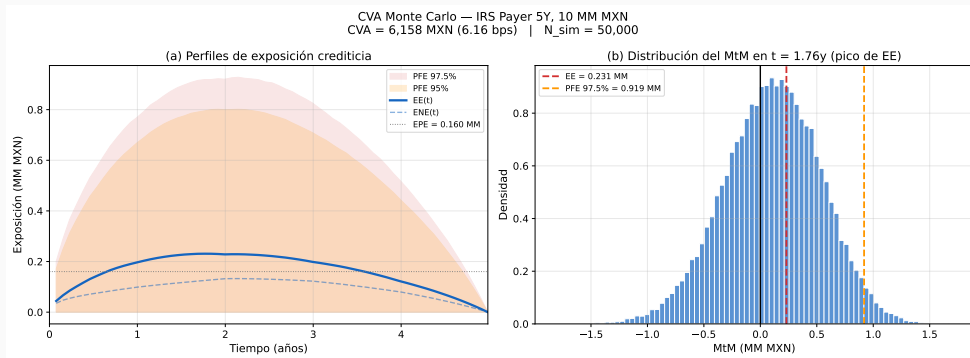


PCA on the nodes up to 5 years, where the swap has sensitivity

10,000 possible futures of TIE de Fondeo



The exposure profile: the hump



It rises through rate diffusion and falls as fewer cash flows remain (amortisation); the peak arrives near $T/3$ (Gregory, 2012). Illustrative figures.

The price of counterparty risk

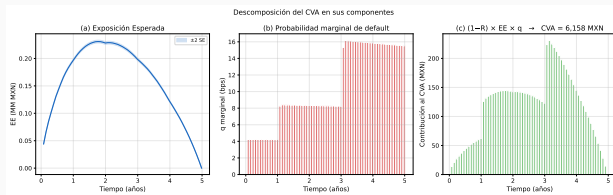
$$\text{CVA} = (1 - R) \int_0^T \text{DF} \text{EE} d\text{PD}$$

on the order of 6 bps

(illustrative)

The hump $\times$ the probability: the integrand shows *in which periods* the risk concentrates. That is the discount the desk builds into the quote, today.

Validating the engine matters: leaving out the coupon accrued between payment dates can move a book's CVA by about 10%.



The same reference *swap*: exposure, marginal default probability and each period's contribution, undiscounted (the 6,158 MXN CVA includes discounting).

From one swap to the book: the operational problem

A calculation of tenths of a second per *swap* sounds fast...until you multiply it:

- A book of 795 *swaps* $\Rightarrow$ minutes per full valuation.
- $\times 40$ bumps for sensitivities $\Rightarrow$ hours.
- $\times 1,000$ stress scenarios $\Rightarrow$ days of compute.

And the desk does not only file the report: **it needs answers during the day.**

Periodic (regulatory)

- NIF B-17 and C-10 (CUB, Annex 33, criterion A-2)
- Banxico XVA report: monthly, by counterparty or by trade type

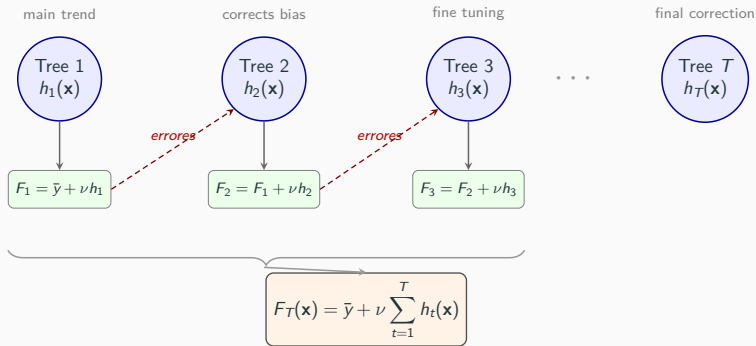
Intraday (operational)

- Quoting: how much CVA does *this* trade add? (Circular 4/2012, Annex 1)
- Sensitivities and hedging the book
- Where does the CVA come from?
Trade-level attribution

Machine learning: a surrogate trained on the engine

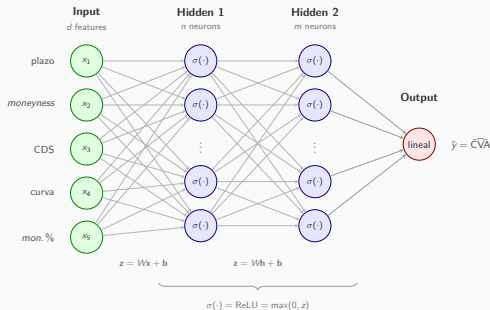
1. The Monte Carlo engine generates the training data: pairs (swap, market) $\rightarrow$ CVA. Labels come from simulation, with no manual labelling.
2. A supervised model learns that function. Two families are compared: boosted decision trees (XGBoost) and neural networks.
3. Once trained, the surrogate evaluates CVA at a much lower cost than the full engine, especially in batch. The engine remains the reference for reporting and for validating the surrogate.

XGBoost: hundreds of trees correcting one another



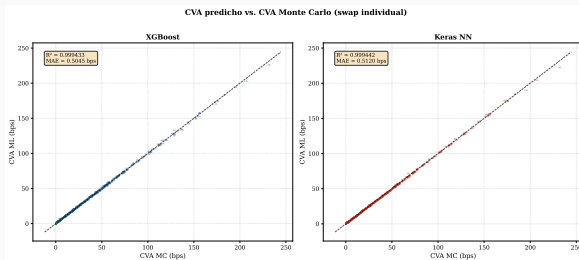
Each tree corrects the errors of the previous one (*boosting*); in the thesis, the mean error on the CVA level is a fraction of a basis point.

The neural network: a composition of simple functions



It is trained by adjusting weights until the validation error stops falling; its derivatives come from automatic differentiation (derivatives of the surrogate, not of the engine).

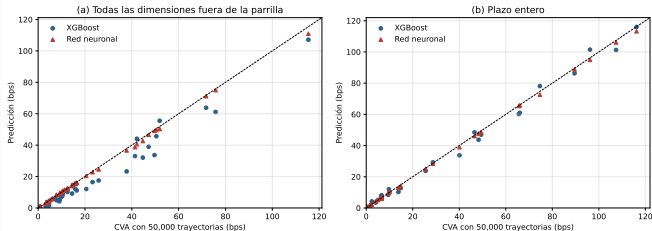
How well does it learn? $R^2 > 0.999$



	Monte Carlo	ML surrogates
Time per <i>swap</i>	tenths of a second	μs per row (in batch)
Batch speed-up	$1\times$	10^3 to $10^5\times$

Thesis wall-clock times, on a single machine and in its configuration: average per query when pricing in batch (an isolated query takes longer). Training is excluded. The engine generates the labels, validates the surrogate and keeps the auditable report.

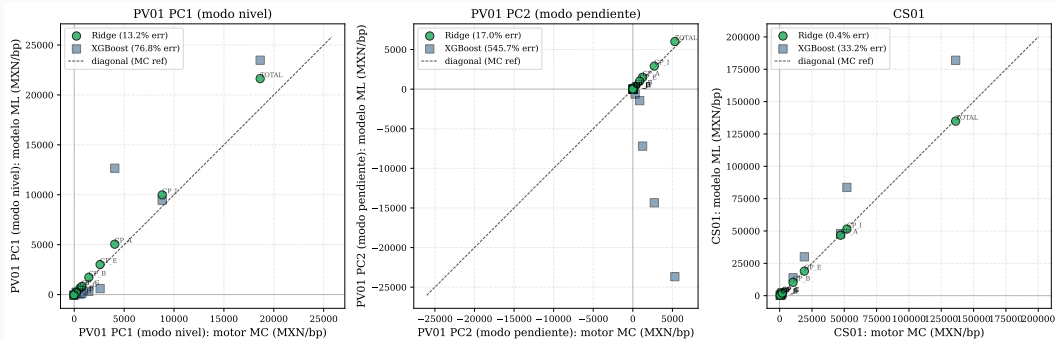
And on values between the grid nodes?



MAE (bps)	XGBoost	Neural network
On the grid nodes	0.50	0.55
All variables between nodes	4.43	0.66

XGBoost is piecewise constant and saw only ten tensors; the network is continuous. Test values lie within the training ranges (interpolation), with labels from 50,000 paths. The network was retrained for this test (0.55 bps on the nodes, versus 0.51 on the previous slide).

Predicting CVA is not the same as predicting its derivatives



Differentiating the XGBoost model trained on levels gives errors from 33.2% (CS01) to 545.7% (PC2 PV01, with the wrong sign). For sensitivities, the thesis uses Ridge regression: the figure shows the global version; the local version cuts the error to 0.15%–2.31% within its domain.

In the *notebook*, in this order:

1. Simulate 10,000 G2++ paths.
2. The distribution split at zero: where CVA and DVA come from.
3. CVA, DVA and the bilateral adjustment, with our own *spread*.
4. The race: full engine, engine with a stored cube, and surrogate.
5. Quote five alternatives and reconcile them against the engine.

What we saw, by area of study

- Stochastic processes and the measure $\mathbb{Q}$ → the G2++ and the simulation.
- Numerical methods → Monte Carlo, *bootstrap*, calibration.
- *Machine learning* → the surrogate pricer and its validation.
- Credit and counterparty risk → CVA, DVA, exposure.
- Intraday portfolio management → *netting*, incremental *pre-deal* CVA, sensitivities and trade-level attribution.
- Mexican regulation → CNBV, Banco de Mexico, the XVA report.



The complete *notebook* for this class:

- G2++ simulation and *swap* valuation
- CVA, DVA and the bilateral adjustment
- The *machine learning* surrogate

Runs in the browser with Google Colab (nothing to install) or on your machine with `pip install -r requirements.txt`.

Questions?

Market data: Valmer (curves and CDS), Bloomberg (*swaptions*)